

Examples

This chapter shows three examples, first a trivial one, next an advanced one and finally one where the specified costs are used. Each subsection starts with a quick description of the example, followed by a listing of the settings and goals and ends with the grid and command output.

Two Entrances

The first example is a room with two doors, where only one door is open, never both, never none. This is a simple one, which can be done manually very quickly, by anybody who is familiar with redstone circuits.

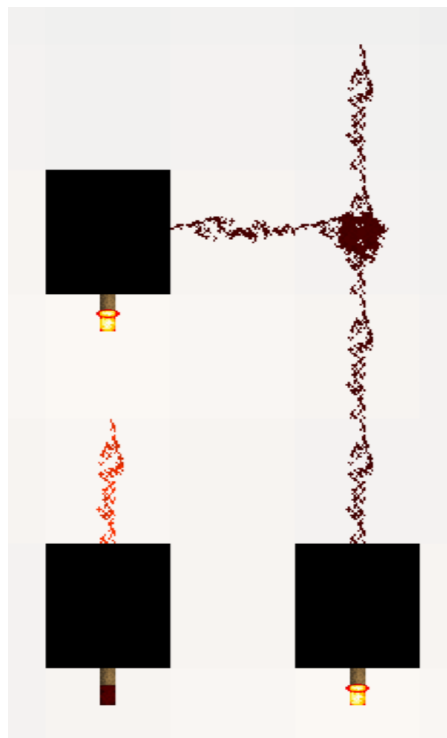


Figure 1: example: two entrances

Settings:

(define nCOLs 3)
(define nROWS 6)

Goals:

(set_input "rs" 0 15 2 0)
(add_off_on_torch_goal 0 5)
(add_on_off_torch_goal 2 5)

Grids:

BL	0	1	2
0	ai	ai	rs
1	so	rs	rs
2	ts	ai	rs
3	rs	ai	rs
4	so	ai	so
5	ts	ai	ts

IR	0	1	2	SF	0	1	2
0	0	0	0	0	0	0	15
1	0	0	0	1	13	13	14
2	16	0	0	2	0	0	13
3	15	0	0	3	0	0	12
4	15	0	0	4	0	0	12
5	0	0	16	5	16	0	0

IF	0	1	2	SF	0	1	2
0			.	0			*
1	O	.	.	1	O	.	*
2	v		.	2	!v		*
3	*		.	3	.		*
4	O		O	4	O		O
5	!v		v	5	v		!v

Command to set the blocks:

```
/summon falling_block ~ ~ 1 ~ {BlockState:{Name:redstone_block},Passengers:[
{id:armor_stand,Health:0,Passengers:[
{id:falling_block,BlockState:{Name:activator_rail},Passengers:[
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~0 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'setblock ~0 ~-2 ~1 stone'},
{id:command_block_minecart,Command:'setblock ~1 ~-2 ~1 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~1 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'setblock ~0 ~-2 ~2
redstone_wall_torch[facing=south]'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~2 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'setblock ~0 ~-2 ~3 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~3 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'setblock ~0 ~-2 ~4 stone'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~4 stone'},
{id:command_block_minecart,Command:'setblock ~0 ~-2 ~5
redstone_wall_torch[facing=south,lit=false]'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~5
redstone_wall_torch[facing=south]'},
{id:command_block_minecart,Command:'summon falling_block ~ ~ -1.5 ~
{BlockState:{Name:command_block}}'},
{id:command_block_minecart,Command:'setblock ~ ~ 1 ~ command_block
{auto:1,Command:"fill ~ ~ ~ ~ -3 ~ air"}'},
{id:command_block_minecart,Command:
'kill @e[type=command_block_minecart,distance=..1]'}]]]]}]}
```

Command to clear the blocks:

```
/summon falling_block ~ ~ 1 ~ {BlockState:{Name:redstone_block},Passengers:[
{id:armor_stand,Health:0,Passengers:[
{id:falling_block,BlockState:{Name:activator_rail},Passengers:[
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~0 air'},
{id:command_block_minecart,Command:'setblock ~0 ~-2 ~1 air'},
{id:command_block_minecart,Command:'setblock ~1 ~-2 ~1 air'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~1 air'},
{id:command_block_minecart,Command:'setblock ~0 ~-2 ~2 air'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~2 air'},
{id:command_block_minecart,Command:'setblock ~0 ~-2 ~3 air'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~3 air'},
{id:command_block_minecart,Command:'setblock ~0 ~-2 ~4 air'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~4 air'},
{id:command_block_minecart,Command:'setblock ~0 ~-2 ~5 air'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~5 air'},
{id:command_block_minecart,Command:'summon falling_block ~ ~ -1.5 ~
{BlockState:{Name:command_block}}'},
{id:command_block_minecart,Command:'setblock ~ ~ 1 ~ command_block
{auto:1,Command:"fill ~ ~ ~ ~ -3 ~ air"}'},
{id:command_block_minecart,Command:
'kill @e[type=command_block_minecart,distance=..1]'}]]]]}]}
```

Energy Saving

The second example represents an energy saving mode of a room, as is either lit by four lamps or by only one. Trying to find a solution for this problem can be very tricky, even for redstone experts. This is not, because of the logic behind the circuit being inherently complex. It is because of the limited space and therefore tightly packed circuit, so that even understanding the proposed result of the algorithm takes a few moments.

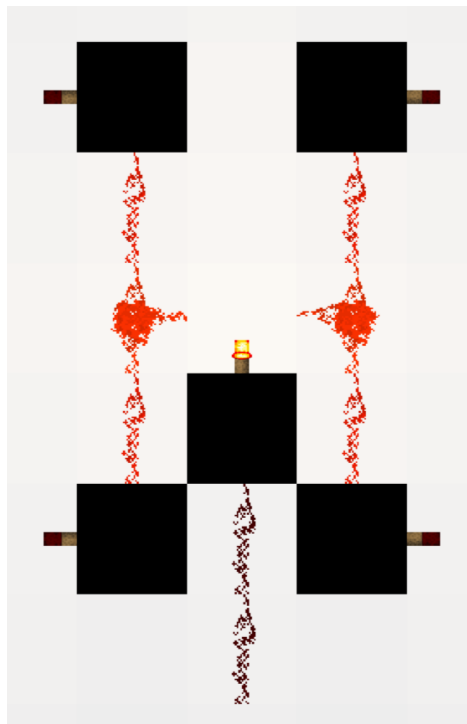


Figure 2: example: energy saving

Settings:

```
(define nCOLs 5)
(define nROWS 6)
```

Goals:

```
(set_input "rs" 0 15 2 5)
(add_on_off_torch_goal 2 2)
(add_off_on_torch_goal 0 0)
(add_off_on_torch_goal 4 0)
(add_off_on_torch_goal 0 4)
(add_off_on_torch_goal 4 4)
```

Grids:

BL	0	1	2	3	4
0	tw	so	ai	so	te
1	ai	rs	ai	rs	ai
2	ai	rs	tn	rs	ai
3	ai	rs	so	rs	ai
4	tw	so	rs	so	te
5	ai	ai	rs	ai	ai

IR	0	1	2	3	4	SR	0	1	2	3	4
0	0	14	0	14	0	0	16	0	0	0	16
1	0	14	0	14	0	1	0	0	0	0	0
2	0	15	16	15	0	2	0	0	0	0	0
3	0	14	0	14	0	3	0	0	14	0	0
4	0	14	0	14	0	4	16	0	14	0	16
5	0	0	0	0	0	5	0	0	15	0	0

IF	0	1	2	3	4	SF	0	1	2	3	4
0	! $\lt$	O		O	! $\gt$	0	$\lt$	0		0	$\gt$
1		*		*		1		.		.	
2		*	$\wedge$	*		2		.	! $\wedge$	.	
3		*	O	*		3		.	0	.	
4	! $\lt$	O	.	O	! $\gt$	4	$\lt$	0	*	0	$\gt$
5			.			5			*		

Command to set the blocks:

```

/sound falling_block ~ ~ 1 ~ {BlockState:{Name:redstone_block},Passengers:[
  {id:armor_stand,Health:0,Passengers:[
    {id:falling_block,BlockState:{Name:activator_rail},Passengers:[
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~0
        redstone_wall_torch[facing=west,lit=false]'},
      {id:command_block_minecart,Command:'setblock ~1 ~-2 ~0 stone'},
      {id:command_block_minecart,Command:'setblock ~3 ~-2 ~0 stone'},
      {id:command_block_minecart,Command:'setblock ~4 ~-2 ~0
        redstone_wall_torch[facing=east,lit=false]'},

```

```

{id:command_block_minecart,Command:'setblock ~1 ~-2 ~1 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'setblock ~3 ~-2 ~1 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'setblock ~1 ~-2 ~2 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~2
    redstone_wall_torch[facing=north]'},
{id:command_block_minecart,Command:'setblock ~3 ~-2 ~2 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'setblock ~1 ~-2 ~3 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~3 stone'},
{id:command_block_minecart,Command:'setblock ~3 ~-2 ~3 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'setblock ~0 ~-2 ~4
    redstone_wall_torch[facing=west,lit=false]'},
{id:command_block_minecart,Command:'setblock ~1 ~-2 ~4 stone'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~4 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'setblock ~3 ~-2 ~4 stone'},
{id:command_block_minecart,Command:'setblock ~4 ~-2 ~4
    redstone_wall_torch[facing=east,lit=false]'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~5 redstone_wire[west=up]'},
{id:command_block_minecart,Command:'summon falling_block ~ ~-1.5 ~
    {BlockState:{Name:command_block}}'},
{id:command_block_minecart,Command:'setblock ~ ~1 ~ command_block
    {auto:1,Command:"fill ~ ~ ~ ~-3 ~ air"}'},
{id:command_block_minecart,Command:
    'kill @e[type=command_block_minecart,distance=..1]'}]]]]}

```

Command to clear the blocks:

```

/summon falling_block ~ ~1 ~{BlockState:{Name:redstone_block},Passengers:[
{id:armor_stand,Health:0,Passengers:[
{id:falling_block,BlockState:{Name:activator_rail},Passengers:[
{id:command_block_minecart,Command:'setblock ~0 ~-2 ~0 air'},
{id:command_block_minecart,Command:'setblock ~1 ~-2 ~0 air'},
{id:command_block_minecart,Command:'setblock ~3 ~-2 ~0 air'},
{id:command_block_minecart,Command:'setblock ~4 ~-2 ~0 air'},
{id:command_block_minecart,Command:'setblock ~1 ~-2 ~1 air'},
{id:command_block_minecart,Command:'setblock ~3 ~-2 ~1 air'},
{id:command_block_minecart,Command:'setblock ~1 ~-2 ~2 air'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~2 air'},
{id:command_block_minecart,Command:'setblock ~3 ~-2 ~2 air'},
{id:command_block_minecart,Command:'setblock ~1 ~-2 ~3 air'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~3 air'},
{id:command_block_minecart,Command:'setblock ~3 ~-2 ~3 air'},
{id:command_block_minecart,Command:'setblock ~0 ~-2 ~4 air'},
{id:command_block_minecart,Command:'setblock ~1 ~-2 ~4 air'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~4 air'},
{id:command_block_minecart,Command:'setblock ~3 ~-2 ~4 air'},
{id:command_block_minecart,Command:'setblock ~4 ~-2 ~4 air'},
{id:command_block_minecart,Command:'setblock ~2 ~-2 ~5 air'},
{id:command_block_minecart,Command:'summon falling_block ~ ~-1.5 ~
    {BlockState:{Name:command_block}}'},
{id:command_block_minecart,Command:'setblock ~ ~1 ~ command_block
    {auto:1,Command:"fill ~ ~ ~ ~-3 ~ air"}'},
{id:command_block_minecart,Command:
    'kill @e[type=command_block_minecart,distance=..1]'}]]]]}

```

Expensive Redstone

The final example demonstrates the effect of using specified costs. Redstone dust has a default value of two in the list `SPECIFIED_COSTS`, as it is fairly easy to get. In this example, the synthesis is done two times, first with the standard value and next with an overly high price of nine. This shows that the algorithm instantly adapts and finds the cheapest solution for the new settings.



Figure 3: example: expensive redstone dust

Settings:

```
(define nCOLs 1)
(define nROWS 12)
(define USE_SPECIFIED_COSTS #f)
SPECIFIED_COSTS: redstone dust = 2 or 9
```

Goals:

```
(add_str_goal = 0 0 11)
(add_str_goal > 0 0 11 #f)
```

Grids (cheap redstone dust):

BL	0	IR	0	IF	0	SR	0	SF	0
0	rs	0	0	0	.	0	15	0	*
1	rs	1	0	1	.	1	14	1	*
2	rs	2	0	2	.	2	13	2	*
3	rs	3	0	3	.	3	12	3	*
4	rs	4	0	4	O	4	11	4	O
5	rs	5	0	5	v	5	10	5	!v
6	rs	6	0	6	*	6	9	6	.
7	rs	7	0	7	*	7	8	7	.
8	rs	8	0	8	*	8	7	8	.
9	rs	9	0	9	*	9	6	9	.
10	rs	10	0	10	O	10	5	10	O
11	so	11	0	11	!v	11	5	11	v

Grids (expensive redstone dust):

BL	0	IR	0	IF	0	SR	0	SF	0
0	rs	0	0	0	.	0	15	0	*
1	so	1	0	1	O	1	15	1	O
2	ts	2	16	2	v	2	0	2	!v
3	rs	3	15	3	*	3	0	3	.
4	so	4	15	4	O	4	0	4	O
5	ts	5	0	5	!v	5	16	5	v
6	rs	6	0	6	.	6	15	6	*
7	so	7	0	7	O	7	15	7	O
8	ts	8	16	8	v	8	0	8	!v
9	rs	9	15	9	*	9	0	9	.
10	so	10	15	10	O	10	0	10	O
11	ts	11	0	11	!v	11	16	11	v

Command to set the blocks (cheap redstone dust):

```
/summon falling_block ~ ~1 ~ {BlockState:{Name:redstone_block},Passengers:[
  {id:armor_stand,Health:0,Passengers:[
    {id:falling_block,BlockState:{Name:activator_rail},Passengers:[
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~0 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~1 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~2 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~3 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~4 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~5 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~6 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~7 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~8 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~9 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~10 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~11 stone'},
      {id:command_block_minecart,Command:'summon falling_block ~ ~-1.5 ~
        {BlockState:{Name:command_block}}'},
      {id:command_block_minecart,Command:'setblock ~ ~1 ~ command_block
        {auto:1,Command:"fill ~ ~ ~ ~-3 ~ air"}'},
      {id:command_block_minecart,Command:
        'kill @e[type=command_block_minecart,distance=..1]'}]]]]}]}
```

Command to set the blocks (expensive redstone dust):

```
/summon falling_block ~ ~1 ~ {BlockState:{Name:redstone_block},Passengers:[
  {id:armor_stand,Health:0,Passengers:[
    {id:falling_block,BlockState:{Name:activator_rail},Passengers:[
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~0 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~1 stone'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~2
        redstone_wall_torch[facing=south]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~3 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~4 stone'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~5
        redstone_wall_torch[facing=south,lit=false]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~6 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~7 stone'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~8
        redstone_wall_torch[facing=south]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~9 redstone_wire[west=up]'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~10 stone'},
      {id:command_block_minecart,Command:'summon falling_block ~ ~-1.5 ~
        {BlockState:{Name:command_block}}'},
      {id:command_block_minecart,Command:'setblock ~ ~1 ~ command_block
        {auto:1,Command:"fill ~ ~ ~ ~-3 ~ air"}'},
      {id:command_block_minecart,Command:
        'kill @e[type=command_block_minecart,distance=..1]'}]]]]}]}
```

Command to clear the blocks:

```
/summon falling_block ~ ~1 ~ {BlockState:{Name:redstone_block},Passengers:[
  {id:armor_stand,Health:0,Passengers:[
    {id:falling_block,BlockState:{Name:activator_rail},Passengers:[
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~0 air'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~1 air'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~2 air'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~3 air'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~4 air'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~5 air'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~6 air'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~7 air'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~8 air'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~9 air'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~10 air'},
      {id:command_block_minecart,Command:'setblock ~0 ~-2 ~11 air'},
      {id:command_block_minecart,Command:'summon falling_block ~ ~-1.5 ~
        {BlockState:{Name:command_block}}'},
      {id:command_block_minecart,Command:'setblock ~ ~1 ~ command_block
        {auto:1,Command:"fill ~ ~ ~ ~-3 ~ air"}'},
      {id:command_block_minecart,Command:
        'kill @e[type=command_block_minecart,distance=..1]'}}]]]]}
```